

Given: $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 0 & 1 \end{bmatrix}$, $b = [2 \ 1 \ -2]^T$, $A^T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

1.) Is $Ax = b$ consistent?:

Row 1: $x_1 + x_2 = 2$

So, $x_1 + (-2) = 2$

Then, $4 + 2(-2) = 1$

Row 2: $x_1 + 2x_2 = 1$

$x_1 = 4$

$0 \neq 1$

$x_2 = -2$

So, the system is inconsistent.

$$2) A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+1+0 & 1+2+0 \\ 1+2+0 & 1+4+1 \\ 1+2+0 & 1+4+1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$$

$$3.) A^T b = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 2+1+0 \\ 2+2+(-2) \\ 2+2+(-2) \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}$$

$$4.) \text{Solve } A^T A x = A^T b: \begin{bmatrix} 2 & 3 & | & 3 \\ 3 & 6 & | & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & | & 4 \\ 0 & 1 & | & -5/3 \end{bmatrix}$$

$$\text{So } x = \begin{bmatrix} 4 \\ -5/3 \end{bmatrix}$$

5.) Compute the orthogonal projection $\text{Proj}_S b = Ax$:

$$\text{Proj}_S b = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ -5/3 \end{bmatrix} = \begin{bmatrix} 4 - 5/3 \\ 4 - 10/3 \\ 0 - 5/3 \end{bmatrix} = \begin{bmatrix} 7/3 \\ 2/3 \\ -5/3 \end{bmatrix}$$